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The Impact of Discrimination and Poor Provider Communication on Public Trust in Cancer Advice From Physicians

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Abstract

Trust in physicians is an important factor influencing patient engagement and health outcomes. While prior studies have explored the role of trust in cancer patients, there is limited research assessing trust in physicians regarding cancer-related information, including prevention, among the general population. This study aims to assess the impact of negative healthcare experiences, including poor provider communication and discrimination, on physician trust in cancer-related advice. This study utilized the Health Information National Trends Survey 6, a nationally representative survey. Binary logistic regression was performed to identify associations between covariates and trust in physicians' cancer information. Patients reporting dissatisfaction across all 7 patient-communication categories exhibited the lowest rates of trust among all covariates (aOR: 0.34, CI [0.28-0.42]). Respondents who experienced racial or ethnic discrimination in healthcare were less likely to express high trust (aOR: 0.65, CI [0.50-0.84]). Racial disparities were also identified, with non-Whites and Hispanics reporting lower levels of trust. This study highlights the significant association of negative healthcare interactions on trust in physicians' cancer-related advice and underscores the need for improved patient-centered communication.

Keywords

physician trust, medical mistrust, healthcare disparities, cancer prevention, cancer screening, physician–patient relationship, physician–patient communication, medical practice

Introduction

Trust between a physician and their patient is an important component of the physician–patient relationship.[1,2] However, trust in physicians and hospitals has decreased substantially from 72% in April 2020 to 40% in January 2024, likely due in part to the COVID-19 pandemic.[3] A better understanding of the current landscape of physician trust is crucial, as it has an impact on patient health. Overall, patient trust has been shown to increase willingness to see a physician,[4] and one meta-analysis found that trust in one's healthcare provider is correlated with improved health outcomes.[5]

Among cancer patients, there are unique factors that contribute to a patient's trust. One review identified that perceived technical competence, honesty, and patient-centered behavior by physicians enhanced a cancer patient's trust in their physician.[6] A recent study in breast cancer patients found that trust in healthcare providers was found to be associated with general satisfaction, technical quality, interpersonal manner, and communication.[7] Patient anxiety is an additional factor associated with decreased trust in a cancer patient's radiation oncologist.[8] Rates of trust among cancer patients also appear to differ among races; for example, Black women with breast cancer report much lower in healthcare providers' trust than White women with breast cancer.[7] Trust remains crucial among those with a cancer history, as among cancer survivors, trust in a follow-up care physician was associated with better health-related quality of life.[9]

However, a patient's trust in cancer-related information and advice from their primary care physician, or other physicians, remains important regardless of the presence of a cancer diagnosis. For example, participation in cancer

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detection measures, such as screening, is influenced by one's perception and trust of their physician. Trust in primary care physicians was found to increase the rate of colorectal cancer screening.[10] Conversely, healthcare distrust was associated with lower breast and cervical cancer screening rates.[11] In a qualitative analysis, it emerged that patient-centered communication and encouragement influenced trust in physicians and the medical system. The opposite was also found to be true, and a lack of patient-centered communication and encouragement, as well as perceptions of poor quality of care, led to medical mistrust.[12]

Overall, the relationship between trust, patient behavior, and outcomes is fairly well documented. However, factors that may influence one's trust in physicians, specifically regarding trust in cancer-related information, have not yet been established in a quantitative study. This study aims to assess the association of a negative healthcare experience due to poor provider communication or discrimination on one's trust in their physician regarding cancer advice. Even among those who do not have cancer, it is still crucial to trust a physician's guidance regarding cancer, such as regarding screening. This study will guide on the association between poor patient communication, as well as discrimination, on a patient's trust in cancer-related information from a physician.

Methods

Survey Design

Data for this study were obtained from the Health Information National Trends Survey 6 (HINTS-6) database.[13] HINTS is sponsored and conducted by the National Cancer Institute and asks questions that focus on cancer beliefs, health behaviors and beliefs, and more.[14] HINTS is administered using equal probability sampling to random samples of adults aged 18 or older residing in the United States. The survey is conducted by mail and online in English and Spanish with a compensation of $2 as an incentive for completion.[15]

Before HINTS-6, HINTS-5 was administered from 2017 to 2020.[16] Trust in physicians and healthcare has decreased significantly compared to before the COVID-19 pandemic in 2020.[3] Due to the unique status of healthcare trust post-COVID-19, only data from HINTS-6 was analyzed. All HINTS data is previously deidentified before being made public, and therefore, IRB approval was not required.

Independent and Dependent Variables

Physician trust in cancer information was assessed using the following question: In general, how much would you trust information about cancer from a doctor (A lot, Some, A little, Not at all)? Patient experiences with healthcare were assessed via their self-reported history of communication and discrimination. Communication was assessed using the following questions: About your communication with all doctors, nurses, or other health professionals you saw during the past 12 months. How often did they do each of the following (Always, Usually, Sometimes, Never)? (1) Give you the chance to ask all the health-related questions you had (2) Give the attention you needed to your feelings and emotions (3) Involve you in decisions about your health care as much as you wanted (4) Make sure you understood the things you needed to do to take care of your health (5) Explain things in a way you could understand (6) Spend enough time with you (7) Help you deal with feelings of uncertainty about your health or health care. In the regression analysis, these 7 questions were combined into a composite variable ranging from 0 to 7 by summing responses to the seven survey items, with 0 representing a response of "Never" in all categories and 7 representing a response of "Always," "Usually," or "Sometimes" to all 7 categories. An additional question was considered for patient perceptions of racial or ethnic discrimination: Have you ever been treated unfairly or been discriminated against when getting medical care because of your race or ethnicity (yes or no)?

Many demographic covariates and confounding variables were included in the analyses. Demographic variables included current age, sex at birth (male or female), race (White, Black, multiracial, or other), ethnicity (Hispanic/Latino or not), income ($0-$14,999, $15,000-$49,999, $50,000-$99,999, or ≥ $100,000), education (non-high school graduate, high school graduate, or college graduate), marital status (single, married or with partner, divorced or separated, or widowed), current employment status (employed or not employed), has insurance (yes or no), lives in rural area (yes or no). Additional health information was collected such as prior cancer diagnosis (yes or no), family member prior diagnosis (yes or no), currently a cancer caregiver (yes or no), currently a caregiver for anything (yes or no), health status (excellent/very good or not), and interest in screening (not interested, somewhat interested, very interested/up to date).

Statistical Analysis

Descriptive and bivariate analyses were used in this study and included the mean and standard deviation for continuous variables and proportions for categorical variables. The Chi-squared test and ANOVA were used for the statistical comparison of discrete and continuous variables, respectively. Binary logistic regression analyses were used to assess the association between covariates and having a lot of trust in cancer information from a physician. All variables in the bivariate analysis with a $P<.10$ were included in the regression. All statistical tests were two-tailed, with a $P<.05$ being considered significant. Statistical analyses were performed using SPSS statistical software version 29.0.

Results

Overall, 6158 participants responded to the question: "In general, how much would you trust information about

cancer from a doctor?" 4468 (72.6%) responded "a lot," 1349 (21.9%) responded "some," 252 (4.1%) responded "a little," and 89 (1.4%) responded "not at all."

Table 1 demonstrates the demographic data of the participants stratified by trust in cancer information from a physician. The mean age of respondents was 55.6 (*SD*: 17.4) and did not significantly differ in different trust level groups ($P = .507$). Trust significantly varied based on income, for example, 12.6% of those with the lowest income (0-$14,999 per year) trusted physicians a little or not at all, whereas only 2.4% of those with the highest income ($100,000+) trusted physicians a little or not at all ($P < .001$). A similar association was seen in education, where 61.9% of non-high school graduates trusted physicians a lot compared to 77.2% of college graduates ($P < .001$). No difference was seen between men and women regarding trust ($P = .180$). Trust significantly varied between racial groups, with White respondents reporting a lot of trust most frequently at 75.3% compared to 69.7% of Black respondents, 63.8% of multiracial respondents, and 66.9% of other races ($P < .001$). Those with health insurance ($P < .001$) and those self-reporting excellent health ($P < .001$) were also significantly more likely to express a lot of trust in a physician's cancer-related advice.

Table 2 demonstrates the cancer history and medical experience of the participants stratified by trust in cancer information from a physician. Having a lot of trust in a physician's cancer information was decreased in those without a prior cancer diagnosis by 9.0% ($P < .001$) and in those without a familial cancer diagnosis by 6.2% ($P < .001$). Notably, caregivers for anything are slightly less likely to trust cancer advice from physicians ($P < .05$), and no difference was found between non caregivers for cancer and cancer caregivers ($P = .986$). Recent negative experiences with a healthcare provider were very highly associated with decreased trust. For patients who had within 12 months not always been given the chance to ask health-related questions, there was a 19.2% decrease in the likelihood of trusting a physician a lot ($P < .001$). For patients who had within 12 months felt attention was not given to their feelings, there was a 14.6% decrease in the likelihood of trusting a physician a lot ($P < .001$). For patients who had within 12 months not felt involved in decision making, there was an 18.3% decrease in the likelihood of trusting a physician a lot ($P < .001$). For patients who had within 12 months not felt they understood the next steps, there was a 21.0% decrease in the likelihood of trusting a physician a lot ($P < .001$). For patients who had within 12 months not been provided clear explanations, there was a 20.8% decrease in the likelihood of trusting a physician a lot ($P < .001$). For patients who had within 12 months felt they had not spent enough time with their healthcare provider, there was a 13.0% decrease in the likelihood of trusting a physician a lot ($P < .001$). For patients who had within 12 months not felt assisted with feelings of uncertainty, there was an 14.2% decrease in the likelihood of trusting a physician a lot ($P < .001$). Among these variables, there was no significant change in patients not trusting physicians at all. Lastly, among patients who have felt discriminated against due to race or ethnicity previously (not just within 12 months), there was a 21.0% decrease in trusting physicians a lot, the largest difference between groups ($P < .001$).

Figure 1A represents a correlation matrix between trust in cancer-related information from different sources. Trust in a physician was moderately correlated with trust in scientists ($r = 0.40$, $P < .001$) and the government ($r = 0.40$, $P < .001$). The correlation between trust in cancer information from scientists and the government was slightly larger ($r = 0.52$, $P < .001$). Conversely, trust in religious organizations was weakly correlated with trust in family members ($r = 0.31$, $P < .001$) and moderately in charitable organizations ($r = 0.47$, $P < .001$). Trust in charitable organizations was also moderately correlated with trust in the government ($r = 0.44$, $P < .001$) and weakly with trust in scientists ($r = 0.32$, $P < .001$). The data presented suggests two largely independent groups of thought where one determines if they trust systemic healthcare (physicians, scientists, government) and other external sources (religion, family, charities). For example, the determination to trust a physician appears to have little association on the decision to trust a religious organization and vice versa. Figure 1B displays the distribution of the amount of trust for each group. Trust for physicians and scientists appears to be left-skewed, with high levels of trust, while trust for religion appears to be right-skewed. The largest number of respondents stating "a lot" of trust was for physicians at over 70% and the largest number of respondents stating "none at all" of trust was for religious organizations at nearly 40%.

Table 3 shows the binary logistic regression analyses assessing the association between covariates and having a lot of trust in cancer information from a physician. Four thousand nine hundred eighty-two participants responded to all questions and did not have any missing data and were thus included in the regression. The largest predictor of trust was patients reporting poor provider communication, with those who have issues in all 7 categories having 66% lower odds of trusting a lot (aOR: 0.34, CI [0.28-0.42]) and those who have issues in 5 to 6 categories having 43% lower odds of trusting a lot (aOR: 0.57, CI [0.45-0.73]) compared to those not reporting any poor communication. Furthermore, even a single instance of discrimination can decrease the odds of trusting cancer advice from physicians by 35% (aOR: 0.65, CI [0.50-0.84]). Other important significant associations include race, as people who are White were much more likely to trust than people who are Black (aOR: 0.76, CI [0.62-0.94]), multiracial (aOR: 0.66, CI [0.46-0.95]), and other (aOR: 0.73, CI [0.55-0.96]). Those who are Hispanic/Latino were also less likely to trust (aOR: 0.79, CI [0.64-0.97]). Income and education only remained significant in the higher brackets; those with an annual income of $100,000+ were more likely to trust than those making $0-$14,999 (aOR: 1.98, CI: 1.45-2.72) and those with a college education were more likely to trust than those without a high school education (aOR: 1.45, CI

Table 1. Demographic Data of Participants Stratified by Physician Trust.

	A lot	Some	A little	Not at all	*P*
Age, mean (SD)	55.3 (17.6)	55.9 (17.0)	54.3 (16.2)	56.0 (18.8)	.507
Race, *N* (%)					<.001
White	2945 (75.3%)	793 (20.3%)	130 (3.3%)	42 (1.1%)	
Black	708 (69.7%)	243 (23.9%)	49 (4.8%)	16 (1.6%)	
Other	275 (66.9%)	103 (25.1%)	22 (5.4%)	11 (2.7%)	
Multiracial	136 (63.8%)	54 (25.4%)	15 (7.0%)	8 (3.8%)	
Annual income, *N* (%)					<.001
$0-$14 999	430 (63.8%)	159 (23.6%)	52 (7.7%)	33 (4.9%)	
$15 000-$49 999	1166 (68.8%)	419 (24.7%)	79 (4.7%)	30 (1.8%)	
$50 000-$99 999	1187 (73.4%)	365 (22.2%)	59 (3.6%)	7 (0.4%)	
$100 000+	1196 (81.9%)	229 (15.7%)	26 (1.8%)	9 (0.6%)	
Education, *N* (%)					<.001
Non-high school graduate	231 (61.9%)	93 (24.9%)	32 (8.6%)	17 (4.6%)	
High school graduate	1885 (70.1%)	648 (24.1%)	120 (4.5%)	37 (1.4%)	
College graduate	2084 (77.2%)	507 (18.8%)	79 (2.9%)	28 (1.0%)	
Hispanic/Latino, *N* (%)					<.001
Is Hispanic or Latino	661 (67.6%)	245 (25.1%)	50 (5.1%)	22 (2.2%)	
Is not Hispanic or Latino	3401 (74.7%)	943 (20.7%)	159 (3.5%)	51 (1.1%)	
Marital status, *N* (%)					<.001
Single	799 (72.5%)	225 (20.4%)	58 (5.3%)	20 (1.8%)	
Married or living with partner	2250 (76.0%)	597 (20.2%)	83 (2.8%)	30 (1.0%)	
Divorced or separated	704 (66.7%)	275 (26.0%)	56 (5.3%)	21 (2.0%)	
Widowed	440 (70.0%)	151 (24.0%)	29 (4.6%)	9 (1.4%)	
Currently employed, *N* (%)					.051
Yes	2184 (74.0%)	619 (21.0%)	119 (4.0%)	31 (1.0%)	
No	2026 (71.9%)	633 (22.5%)	110 (3.9%)	50 (1.8%)	
Sex, *N* (%)					.180
Male	1691 (74.1%)	466 (20.4%)	89 (3.9%)	36 (1.6%)	
Female	2508 (72.2%)	781 (22.5%)	142 (4.1%)	42 (1.2%)	
Has health insurance, *N* (%)					<.001
Yes	4060 (73.5%)	1184 (21.4%)	212 (3.8%)	70 (1.3%)	
No	328 (64.2%)	131 (25.6%)	35 (6.8%)	17 (3.3%)	
Rural v metro, *N* (%)					<.001
Rural	546 (68.3%)	204 (25.5%)	44 (5.5%)	5 (0.6%)	
Nonrural	3922 (73.2%)	1145 (21.4%)	208 (3.9%)	84 (1.6%)	
Health status					<.001
Poor/fair	713 (66.8%)	258 (24.2%)	79 (7.4%)	18 (1.7%)	
Good/very good	3170 (74.2%)	914 (21.4%)	145 (3.4%)	46 (1.1%)	
Excellent	437 (73.9%)	118 (20.0%)	15 (2.5%)	21 (3.6%)	

* Missing responses not included in count or percentages.
** Responses of "a lot," "some," "a little," and "not at all" were self-reported by participants in response to "In general, how much would you trust information about cancer from a doctor?"

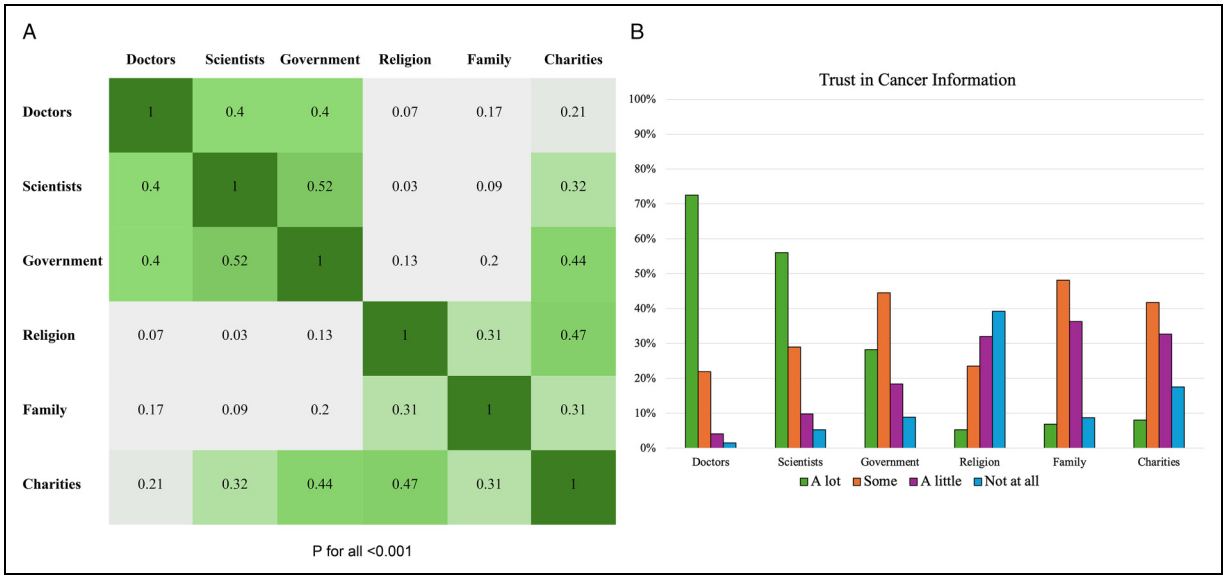

Figure 1. (A) Correlation matrix between sources of cancer information and trust. (B) Frequency distribution of sources of cancer information and trust.

[1.02-2.04]). Additionally, those who are divorced (aOR: 0.70, CI [0.55-0.88]) or widowed (aOR: 0.74, CI [0.55-0.99]) are less likely to trust cancer information from a physician than those who are single.

Supplemental Table S1 displays sensitivity analyses assessing whether the associations shown in the multivariable analysis remained significant in the subgroup of patients. These included only individuals with health insurance and only those who had not previously had cancer. In both models, the results remained the same as the main regression.

Discussion

Our study has identified several variables associated with trust in a physician's cancer-related advice regarding cancer. The main notable finding of our study is that recent negative healthcare communication experiences are strongly associated with medical distrust. Patients responded to 7 categories of prior healthcare experiences within 12 months, and those who responded in the negative to 5 or more categories were significantly less likely to trust a physician. Reacting negatively to all 7 categories was the strongest factor associated with decreased trust for medical advice regarding cancer. This suggests that even a single recent negative experience, if perceived negatively in multiple aspects, can ultimately harm a patient's trust in their provider regarding cancer information. We furthermore found that a single instance of discrimination due to race or ethnicity was significantly associated with decreased trust. These novel findings emphasize that medical professionals must devote time to patient-centered communication and their patients' well-being, as well as avoid actions perceived as derogatory. Additionally, with the sensitivity analyses, we found that among those with health insurance and those who had not previously had cancer, the findings remain consistent, and therefore the findings are applicable to these subpopulations as well.

Our study also found that those who are non-Hispanic White are much more likely to trust cancer advice from physicians than other races. This is consistent with the current literature on overall physician trust. For the general patient population, Black patients specifically are less likely to trust their physician and much more likely to have medical mistrust.[3,17,18] Our findings identify that this disparity remains regarding cancer-related information. Similarly, the same trust disparity has been reported in cancer patients.[7,19] However, one more recent study in Texas found lower levels of medical trust among non-Hispanic White participants with cancer than Black or Hispanic participants.[20]

We identified an association between the lowest and highest levels of income and education and trust, with those with higher education and income trusting more than those at the lowest levels. The association between income and trust has been seen in other studies of healthcare trust in the general population as well.[21] However, conversely to our findings, many studies have found that lower education is associated with an increase in medical trust.[20,22,23] This suggests that medical trust is, in part, context-specific, and individuals may trust physicians differently depending on their health diagnoses and what type of physician is sharing information with them. An additional novel finding of our study is that divorced and widowed individuals are less likely to trust cancer information from a physician. This has not been reported previously and may be due in part to medical biases against divorced patients.[24]

Overall, our study effectively identifies factors associated with trust in cancer-related information from physicians. Research into the various types and influences of medical

Table 2. Cancer and Medical History of Participants Stratified by Physician Trust.

	A lot	Some	A little	Not at all	*P*
Interest in cancer screening, *N* (%)					<.001
Not interested	848 (66.4%)	335 (26.2%)	62 (4.9%)	32 (2.5%)	
A little/somewhat interested	1544 (72.9%)	471 (22.2%)	82 (3.9%)	22 (1.0%)	
Very interested or up to date	1830 (76.2%)	453 (18.9%)	89 (3.7%)	29 (1.2%)	
Current cancer caregiver, *N* (%)					.986
Yes	62 (72.9%)	18 (21.2%)	4 (4.7%)	1 (1.2%)	
No	4238 (72.7%)	1273 (21.8%)	235 (4.0%)	82 (1.4%)	
Current caregiver for anything, *N* (%)					.034
Yes	655 (70.7%)	202 (21.8%)	52 (5.6%)	17 (1.8%)	
No	3664 (73.0%)	1097 (21.9%)	189 (3.8%)	66 (1.3%)	
Knows a family member who had cancer, *N* (%)					<.001
Yes	2935 (74.5%)	824 (20.9%)	140 (3.6%)	41 (1.0%)	
No	1255 (69.9%)	415 (23.1%)	87 (4.8%)	38 (2.1%)	
Previous cancer diagnosis, *N* (%)					<.001
Yes	698 (78.9%)	153 (17.3%)	29 (3.3%)	5 (0.6%)	
No	3524 (71.8%)	1104 (22.5%)	202 (4.1%)	79 (1.6%)	
Within the past 12 months:					
Gave patient chance to ask health-related questions, *N* (%)					<.001
Always	2529 (80.7%)	523 (16.7%)	55 (1.8%)	27 (0.9%)	
Not always	1459 (65.2%)	610 (27.3%)	143 (6.4%)	25 (1.1%)	
Gave attention to feelings and emotions, *N* (%)					<.001
Always	1905 (80.9%)	384 (16.3%)	42 (1.8%)	25 (1.1%)	
Not always	2071 (69.1%)	746 (24.9%)	155 (5.2%)	25 (0.8%)	
Involved patient decision making as wanted, *N* (%)					<.001
Always	2298 (81.3%)	450 (15.9%)	53 (1.9%)	25 (0.9%)	
Not always	1677 (66.4%)	682 (27.0%)	143 (5.7%)	25 (1.0%)	
Ensured patient understood next steps, *N* (%)					<.001
Always	2510 (81.6%)	482 (15.7%)	58 (1.9%)	27 (0.9%)	
Not always	1475 (64.5%)	650 (28.4%)	138 (6.0%)	24 (1.0%)	
Provided clear explanations, *N* (%)					<.001
Always	2580 (81.2%)	507 (16.0%)	61 (1.9%)	28 (0.9%)	
Not always	1406 (64.3%)	623 (28.5%)	135 (6.2%)	22 (1.0%)	
Spent enough time with patient, *N* (%)					<.001
Always	1804 (80.5%)	374 (16.7%)	39 (1.7%)	25 (1.1%)	
Not always	2166 (70.0%)	749 (24.2%)	155 (5.0%)	25 (0.8%)	
Assisted with feelings of uncertainty, *N* (%)					<.001
Always	1746 (81.2%)	347 (16.1%)	36 (1.7%)	21 (1.0%)	
Not always	2208 (69.7%)	774 (24.4%)	161 (5.1%)	27 (0.9%)	
Ever perceived discrimination due to race or ethnicity when getting medical care, *N* (%)					<.001
Felt discriminated against by physicians	275 (56.1%)	155 (31.6%)	47 (9.6%)	13 (2.7%)	
Has not felt discriminated	4096 (74.1%)	1155 (20.9%)	199 (3.6%)	74 (1.3%)	

* Missing responses not included in count or percentages.

** Responses of "a lot," "some," "a little," and "not at all" were self-reported by participants in response to "In general, how much would you trust information about cancer from a doctor?"

trust has only recently begun to be studied. Recently, three unique measures have been developed and validated to test trust in a personal physician, physicians in general, and a healthcare team.[25] Similarly, the trust in oncologist scale (TiOS) was developed for cancer patients, and later, the short form (TiOS-SF).[26,27] However, there is yet to be a direct measure assessing trust in cancer-related information from a nononcologist, such as a primary care physician. Future research assessing public trust in cancer information may give rise to a more in-depth understanding of willingness to screen for cancer or engage in cancer prevention behavior.

Our study identifies prior negative experiences with healthcare communication, including poor communication or discrimination, to be associated with one's trust in a physician's cancer advice and should guide medical professionals in the future on the importance of the physician–patient relationship. In particular, physicians should focus on patient-centered communication, including fostering a relationship, shared decision making, and responding to emotions.[28] These areas of communication have been seen to improve patient recall, understanding, and adherence.[28]

Table 3. Binary Logistic Regression Analyses Demonstrating Associations of Prior Healthcare Experiences and Discrimination With Physician Trust ($n = 4982$).

	aOR	[95% CI]	*P*
Race (White reference)			
Black	0.76	[0.62-0.94]	.010
Other	0.73	[0.55-0.96]	.025
Multiracial	0.66	[0.46-0.95]	.024
Income ($0-$14 999 reference)			
$15 000-$49 999	1.13	[0.88-1.46]	.337
$50 000-$99 999	1.27	[0.97-1.67]	.087
$100 000+	1.98	[1.45-2.72]	<.001
Education (non-high school graduate reference)			
High school graduate	1.18	[0.85-1.63]	.335
College graduate	1.45	[1.02-2.04]	.036
Hispanic or Latino (non-Hispanic/Latino reference)	0.79	[0.64-0.97]	.024
Relationship (single reference)			
Married or living with partner	0.88	[0.71-1.08]	.225
Divorced or separated	0.70	[0.55-0.88]	.003
Widowed	0.74	[0.55-0.99]	.043
Currently employed (nonemployed reference)	0.89	[0.75-1.05]	.172
Has Health Insurance (no health insurance reference)	0.90	[0.66-1.22]	.489
Rural (nonrural reference)	0.83	[0.67-1.02]	.080
Health status (poor/fair reference)			
Good/very good	0.93	[0.76-1.14]	.489
Excellent	0.74	[0.54-1.00]	.050
Cancer screening interest (not interested reference)			
A little/somewhat interested	1.48	[1.21-1.80]	<.001
Very interested or up to date	1.61	[1.33-1.97]	<.001
Caregiver for anything	0.93	[0.77-1.14]	.498
Knows of a family member who had cancer	1.02	[0.86-1.20]	.843
Ever had cancer	1.23	[0.99-1.53]	.068
Poor healthcare experiences prior 12 months (0 reference)			
1-2	1.05	[0.82-1.36]	.692
3-4	0.92	[0.71-1.19]	.510
5-6	0.57	[0.45-0.73]	<.001
7	0.34	[0.28-0.42]	<.001
Felt Discriminated due to race or ethnicity (did not feel discriminate reference)	0.65	[0.50-0.84]	.001

* aOR, adjusted odds ratio; CI, confidence intervals.
** Poor healthcare experience is a composite variable ranging from 0 to 7. It was calculated by summing responses to the 7 survey items regarding healthcare experience. 0 represents a response of "Never" in all categories and 7 representing a response of "Always," "Usually," or "Sometimes" to all 7 categories.

Ultimately, while there is an effort to improve access to cancer screening, through programs such as the NCI Cancer Moonshot ACCSIS initiative,[29] there is an additional need to improve cancer information. Implementation of benchmarks and measures for trust in cancer information may guide policymakers in where and how to improve screening rates. Additionally, as reported in this study, perceptions of discrimination and poor communication are not equivocal, and future research should consider these 2 distinct categories that are associated with decreased trust.

Our study has some key limitations. There may be a recall bias with the information being self-reported that could not be validated. Additionally, because this was a cross-sectional analysis, we cannot be certain about causation between trust and certain covariates. The study included a single question to measure trust, which is limited based on the availability of HINTS. Additionally, the survey was administered in 2022, and factors related to trust may have changed as the COVID-19 pandemic progressed. The survey also had some partial respondents; however, we assume the missing data is randomly distributed, and due to the low amount of nonresponse in each question, we assume the data remains useful and generalizable. Lastly, it is important to acknowledge that trust in cancer-related advice is likely influenced by broader trust in the healthcare system, which could limit the novelty of these findings. Despite these limitations, our study utilizes a large, nationally representative sample, and our findings offer novel insights by specifically considering trust in cancer information, an area that requires more research.

Conclusion

This study identifies that trust in cancer advice from a physician is associated with prior healthcare experiences. The

findings suggest that even a single recent negative experience may harm a patient's trust in their provider. Furthermore, our findings emphasize that not all healthcare experiences are equivocal, for example, perceptions of discrimination and poor communication are distinct contributors to medical distrust. As initiatives seek to improve access to cancer screening, it is also important to improve information delivery and provider communication. Overall, in order to maintain and improve public trust in cancer-related medical advice, healthcare providers must prioritize patient-centered communication and actively address discrimination in clinical encounters.

Authorship Contribution

Methodology: A.S and M.W.; formal analysis: A.S.; writing—original draft: A.S.; writing—review and editing: M.W.; visualization: A.S.; supervision: M.W.

Data Availability

The HINTS-6 dataset may be accessed online: https://hints.cancer.gov/data/download-data.aspx.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical Approval

Ethical approval is not required as this study includes secondary data that was previously deidentified by the National Cancer Institute. HINTS-6 received a "Not Human Subjects Research" determination from the NIH Office of IRB Operations on August 16, 2021 (iRISs reference number: 562715).

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Supplemental Material

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Statement of Informed Consent

Informed consent for patient information to be published in this article was not obtained because it included secondary data previously collected.

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